WORKING CAPITAL MANAGEMENT AND SHORT-TERM FINANCING

(INTERNATIONAL CASH MANAGEMENT, SOURCE OF SHORT TERM FINANCING, TRADE CREDIT, RECEIVABLES, PAYABLES, INVENTORY, CASH-MANAGEMENT)

WORKING CAPITAL MANAGEMENT AND SHORT-TERM FINANCING

(INTERNATIONAL CASH MANAGEMENT, SOURCE OF SHORT TERM FINANCING, TRADE CREDIT, RECEIVABLES, PAYABLES, INVENTORY, CASH-MANAGEMENT)

Source:

- *International Financial Management*, Madura, Fox (2nd Ed.): Part 5
- OTP-s nő ppt: Short-term financing
- Juhász videók

Working capital management

The management of working capital has a direct influence on the amount and timing of cash flows. This is because Net Working Capital = Current Assets – Current Liabilities (= Long-term Liab. + Equity – Current Liab.)

Current Assets:

- **Account receivable**
- **Inventory**
- **Cash** & equivalents
- Marketable securities

Current Liabilities:

- **Account payables**
- Short-term borrowing
- Wages
- Tax liabilities
- Interest payable

WORKING CAPITAL MANAGEMENT AND SHORT-TERM FINANCING

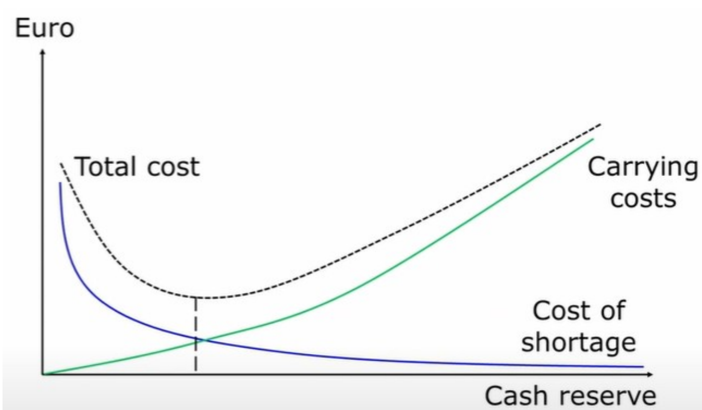
(INTERNATIONAL CASH MANAGEMENT, SOURCE OF SHORT TERM FINANCING, TRADE CREDIT, RECEIVABLES, PAYABLES, INVENTORY, CASH-MANAGEMENT)

The main components of NWC are cash, inventory, receivables and payables. Working capital includes (only) cash that is needed to run the firm on a day-to-day basis. Working capital impacts a firm's value by affecting its FCF.

International cash management

Cash management trade-off

If the company does not have enough cash, it will suffer from paying late its suppliers (worsening relationship), so there is a high *cost of shortage*. The more cash the company keeps, the higher the *carrying costs*, because it cannot earn a return on it. There is an *optimum*: carrying costs imply that investing in the short-term or not investing at all provides little or no return, compared to long-term investment; or it my imply storage costs ex. keeping the cash on a bank account goes with yearly fees. Shortage costs contain costs associated with evading bankruptcy, or explaining suppliers why the company is not paying in time.



Motivation for holding cash:

- To meet day-to-day needs – **transaction balance**
- To compensate for uncertainty associated with CFs (emergency expenses) - **precautionary balance**
- Quick access to liquidity may cost more
- To satisfy bank requirements – **compensating balance** (for services the bank performs)
- Waiting for better investments

Cash cycle

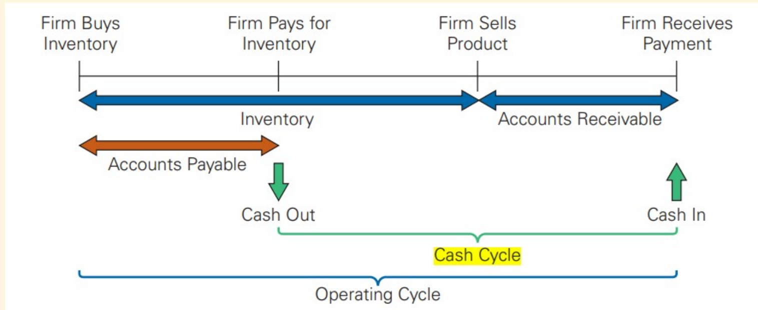
WORKING CAPITAL MANAGEMENT AND SHORT-TERM FINANCING

(INTERNATIONAL CASH MANAGEMENT, SOURCE OF SHORT TERM FINANCING, TRADE CREDIT, RECEIVABLES, PAYABLES, INVENTORY, CASH-MANAGEMENT)

FIGURE 26.1

The Cash and Operating Cycles for a Firm

The cash cycle is the average time between when a firm pays for its inventory and when it receives cash from the sale of its product.



The *operating cycle* contains the acc. rec. period and inventory period: the period from the purchase (w/o payment) of raw materials until cash is collected (this is the longest period). *Cash cycle*, part of operating cycle, is the length of time between when the firm pays cash to purchase its initial inventory and when it receives cash from the sale of the output produced form that inventory. The longer the cash cycle, the more working capital a firm has, and the more cash it needs to conduct its daily operation. Any reduction in WC requirements generates a positive FCF.

Cash Conversion Cycle (CCC) = Acc. Receivable Days + Inventory Days - Acc. Payable Days

Working capital is usually financed through short-term loans from commercial banks. However, *trade credit* is also used: when the firm buys supplies and materials on credit form other firms, it records them as account payables; thus, it is a loan from the selling firm to the buyer, allowing the buyer to pay for the goods at a later time. If the buyer pays earlier than the deadline, it may even get some discount on the payment (ex. 2/10, Net 30: payment is due in 30 days but if paid in 10 days, a 2% discount is applied). A "cash-only" policy may cause the firm to lose its customers due to competition. Trade credit is simple, convenient to use, and has lower transaction costs compared to standard loans.

Centralized cash management

It means that all subsidiary cash positions can be *monitored* and *managed* simultaneously by a **centralized cash management** group. Centralization allows for more efficient usage of funds and possibly lower costs and higher returns. Here the term 'centralized' means that excess cash from each subsidiary is pooled (*cash pooling*) until it is needed by a particular subsidiary.

WORKING CAPITAL MANAGEMENT AND SHORT-TERM FINANCING

(INTERNATIONAL CASH MANAGEMENT, SOURCE OF SHORT TERM FINANCING, TRADE CREDIT, RECEIVABLES, PAYABLES, INVENTORY, CASH-MANAGEMENT)

In case of international companies there may be a **centralized cash management** group which needs to *monitor*, and possibly *manage*, the parent–subsidiary and intersubsidiary cash flows. This group has two tasks:

1. Optimizing cash flow movements
2. Investing excess cash

1. Optimizing cash flow can be done by:
    - Accelerating cash inflows
    - Minimizing currency conversion costs
    - Managing blocked funds
    - Managing intersubsidiary cash transfers

Accelerating cash inflows

The more quickly the inflows are received, the more quickly they can be invested or used for other purposes. A corporation may establish **post office boxes** around the world where customers are instructed to send payment, thus reducing mailing time. Or it can use **preauthorized payments**, which allow a corporation to charge a customer's bank account up to some limit.

Minimizing currency conversion cost

Netting (multilateral netting system) reduces the administrative and transaction costs from currency conversion between subsidiaries/parent company.

Managing blocked funds

Host governments can block funds to create jobs and reduce unemployment. The parent company may instruct the subsidiary to set up an **R&D division**. Alternatively, it could use **transfer pricing**. Or the subsidiary could obtain financing from a local bank instead of the parent.

Managing intersubsidiary cash transfer

If one subsidiary is short of funds, another one could buy supplies from this one, and pay early, thus providing financing. Or the other subsidiary may instead sell supplies to the subsidiary needing finance and allow it to pay later.

There are complications in optimizing CFs:

- Company-related characteristics
- Government restrictions
- Characteristics of banking systems

Company-related characteristics: if one of the subsidiaries delays payments to other subsidiaries for supplies received, the other subsidiaries may be forced to borrow until the payments arrive. A centralized approach should minimize this problem.

Government restrictions: blocking funds, prohibition of netting systems.

Characteristics of banking systems: some services, like lockboxes, are not available in all countries, thus limiting international cash management.

2. Investing excess cash

Eurocurrency deposits are one of the most used international money market instruments. Many MNCs establish large deposits in various currencies in the eurocurrency market, with *eurodollar deposits* being the most popular. MNCs can also purchase *foreign Treasury bills* and *commercial paper*.

If the *two subsidiaries use the same currency*, then if both have excess cash of £50 000 each for one month, the rates on their individual bank deposits may be lower than the rate they could obtain if they pooled their funds into a single £100 000 bank deposit.

If they have funds in multiple currencies, the advantage of pooling may be offset by transaction costs of converting. But it is possible to pool each currency separately, and then use the one in which case another subsidiary has a deficit. Or the money can be invested in securities in the foreign currency that will be needed in the future.

Effective yield

It is the deposit's effective yield, not its interest rate, that is most important to the cash manager. The *effective yield of a bank deposit* considers both the interest rate and the rate of appreciation (or depreciation) of the currency denominating the deposit.

$$r = (1 + i_f)\,(1 + e_f) - 1$$

The effective yield on the foreign deposit is represented by r, i_f is the quoted interest rate, and e_f is the percentage change (from the day of deposit to the day of withdrawal) in the value of the currency representing the foreign deposit.

Financing international trade

In general, five basic methods of payment are used to settle international transactions, each with a different degree of risk to the exporter and importer:

- Prepayment
- Letters of credit (Akkreditív)
- Drafts / Bill of exchange (Váltó)
- Consignment (Bizományi áru)
- Open account

Prepayment

The exporter will not ship the goods until the buyer has remitted payment to the exporter. This method affords the supplier the greatest degree of protection, while the importer relies completely on the exporter to ship the goods.

Letters of credit (L/C)

It is issued by a bank on behalf of the importer (buyer) promising to pay the exporter (beneficiary) upon presentation of shipping documents. The importer pays the issuing bank the amount of the L/C + some fees. The exporter is assured of receiving payment from the issuing bank. The L/C does not guarantee to the importer that the goods purchased will be those invoiced and shipped. It usually involves two banks: the exporter's (the exporter may request a local bank to guarantee the L/C) and the importer's.

Drafts / Bill of exchange

It is an unconditional promise drawn by the exporter, instructing the buyer to pay the face amount of the draft upon presentation. Banks on both ends act as intermediaries. The draft represents the exporter's formal demand for payment from the buyer. If the drawer needs money earlier than the payday, then she can sell it to a bank at a discount. Thus it will become a banker's acceptance (BA).

Consignment

The exporter ships the goods to the importer while still retaining actual title to the goods. Payment happens only after the importer has sold the product. This carries high risk to the exporter (no upfront payment, goods potentially gone) and is limited to affiliated and subsidiary companies. There is no risk to the importer.

Open account transaction

The exporter ships the goods, but the buyer pays only later based on agreed terms. It relies on buyer's creditworthiness and reputation. No risk to the importer. Common among industrialized countries (US, Europe).

Banks on both sides of the transaction play a critical role in financing international trade. The following are some of the more popular methods of financing international trade (not mentioned yet):

- Accounts receivable financing
- Factoring
- Banker's acceptance
- Working capital financing

- Medium-term capital goods financing (forfaiting)
- Countertrade

Accounts receivable financing

Exporters waiting for payment from the importer can get short-term loans from banks using *accounts receivable as collateral.* The *exporter still takes the risk* and responsibility of collecting the payment.

Factoring

Exporters can *sell account receivables* to a third-party (factor) for immediate cash at a discount. The rest is going to be paid to the exporter when payment by customers is made. The factor gets a fee for the process and *takes on the risk* and responsibilities of collecting the payment from the buyer. The factor assesses the importer's creditworthiness, not the exporter's.

Banker's acceptance

A banker's acceptance is a bill of exchange, or time draft, drawn on and accepted by a bank (the importer's bank). It is a short-term guarantee by a bank to pay an exporter (or the holder of the draft) at maturity. Banker's acceptance is a marketable instrument with an active secondary market.

Working capital financing

The bank may even provide short-term loans. The purchase from overseas for the **importer** usually represents the acquisition of inventory. The loan finances the working capital cycle that begins with the purchase of inventory and continues with the sale of the goods, creation of an account receivable, and finally conversion to cash. To **exporters**, their banks can offer short-term loans for pre-export costs (for ex. production costs) or the gap between sale and receiving payment.

Forfaiting

Forfaiting finances expensive machinery or equipment (capital goods) for importers when they cannot make payment for these goods in the short term. The exporter sells debt from the importer (bills or promissory notes) to a forfaiter at a discount. After this, the forfaiter (usually a financial institution) gets paid by the importer over a longer term, assuming the risk instead of the exporter. So, for a fee, the forfaiter enables the exporter to receive the money immediately and the importer at the same time has a period of credit before payment has to be made. This is similar to factoring but for medium-term debt and often with a bank guarantee involved.

Countertrade

The term countertrade denotes all types of foreign trade transactions in which the sale of goods to one country is linked to the purchase or exchange of goods from that same country. Types include barter (straight swap without use of any currency), counterpurchase (separate contracts but linked), and compensation (buyback agreement). Mostly used by governments and large companies, can be complex.

Short-term financing

Short-term financing decisions generally involve short-lived assets and liabilities and are usually easily reversed. The ***matching principle*** states that short-term needs should be financed with short-term debt and long-term needs with long-term sources of funds.

Possible short-term financing methods:

- Syndicated lending
- Commercial bank loans
- Commercial paper
- Account receivable as collateral
- Inventory as collateral

Syndicated lending

Banks engage in syndicated lending to diversify their portfolio. It benefits the borrower in several ways: wide range of maturities, flexibility in loan terms, can be repaid w/o penalty. It usually happens when the borrower is a larger corporation needing a large amount of financing.

Commercial bank loans

The firm pays interest on the loan and pays back the principal in one lump sum at the end of the loan. Fixed or variable interest rate. In a *line of credit,* the bank agrees to lend a firm any amount up to a stated maximum. A *bridge loan* is often used when a firm needs short-term financing until it can arrange for a longer-term financing.

Commercial paper

It is an unsecured debt (bond) issued by large corporations that is usually a cheaper source of funds than a short-term bank loan. It can be *direct paper* (sold directly to investors) or *dealer paper* (dealers sell to investors). Common in the US.

Accounts receivable as collateral

- *Pledging of account receivable*: the lender accepts some of the credit accounts as collateral for the loan, then lends the borrower some percentage of the value of the accepted invoices.
- *Factoring of account receivable*: the firm sells receivables to the lender (factor), and the lender agrees to pay the amount due from the firm's customers.

Inventory as collateral

- In a *floating lien, general lien* or *blanket lien* all the inventory is used to secure the loan. Riskiest to the lender bc the value of inventory changes as it is used/sold by the company.
- In a *trust receipts loan* or *floor planning* the inventory items are held in a trust as security for the loan. If they are sold, the proceeds are used to repay the loan.
- In a *warehouse arrangement*, the inventory serves as collateral for the loan and is stored in a warehouse.

A firm could buy materials on a day-by-day basis, but it would incur higher prices for ordering in small amounts and would also risk *production delays* if the materials are not delivered in time, so ordering more than needed could be a solution. Similarly, a small inventory would lead to *shortage* and may not be sufficient to meet an unexpected increase in demand. There are also cost of holding inventory:

- Acquisition costs (the costs of the inventory itself)
- Order costs (cost of proceeding orders)
- Carrying cost (storage cost, insurance, taxes etc.)

An optimal level of inventory involves a trade-off between *carrying costs* and *order costs*. There are strategies in the manufacturing industry like JIT (just-in-time) in which inventories of parts and subassemblies are nearly zero. This works well when the flow of production in steady and predictable.

Many MNC obtains equity funding in their home country and engage in debt financing in foreign countries. Equity alternatives in the global market are IPO, Euroequity issue, Directed public/private issue, Private placement.

When financing in a foreign currency, it is increasingly the case that the bank or financial market providing the finance will not be in the country of the currency. When this is the case the term ‘euro’ is used. Thus, a eurodollar bank account is an account that will probably be situated in London and will be denominated in dollars. A eurobank is a bank that makes loans and accepts deposits in foreign currencies

Euronotes

They are short term debt securities issued by MNCs w/ 1-6 month maturities. They can be rolled over and they are underwritten by commercial banks.

Euro-commercial paper

Issued by dealers for MNCs w/o the backing of any underwriting syndicate with chosen maturity.

Eurobank loans

Direct loans from eurobanks.

WORKING CAPITAL MANAGEMENT AND SHORT-TERM FINANCING

(INTERNATIONAL CASH MANAGEMENT, SOURCE OF SHORT TERM FINANCING, TRADE CREDIT, RECEIVABLES, PAYABLES, INVENTORY, CASH-MANAGEMENT)

Internal financing

Financing from parent company or from subsidiaries. Sometimes this may be restricted by governments.

Reason of foreign financing

An MNC may finance in a foreign currency to **offset foreign currency inflows** (like net receivable position). Example: Corvinus Ltd (HU) has net receivables denominated in euro and needs forint now for liquidity purposes. It can borrow euros and convert them to forints to obtain the needed funds. Then, the net receivables in euros will be used to pay off the loan.

An MNC may use foreign financing to **reduce costs**. It may be a good decision to borrow in a foreign currency if the interest rates on that currency is relatively low and the currency stable.

Determining the effective financing rate

The actual cost of financing by the debtor firm will depend on:

- the *interest rate charged* by the bank that provided the loan
- the *movement in the borrowed currency's* value over the life of the loan.

When managing international loans, a natural division to examine is how much of the effective interest rate is due to foreign interest rates (i_f) and how much due to the percentage change in the exchange rate (e_f).

$$r_{eff} = (1 + i_f)\,(1 + e_f)$$

Criteria considered for foreign financing

A more formal consideration of the factors relevant to short-term financing should include:

- Interest rate parity
- Exchange rate forecasts

<u>Covered interest arbitrage</u>

Covered interest arbitrage is the process of capitalizing on interest rate differences between two countries while covering exchange rate risk. It differs from conventional arbitrage in that investment and a period of time is required in the process (the funds are tied up for a period of time). If the proceeds from engaging in covered interest arbitrage exceed the proceeds from investing in a domestic bank deposit, covered interest arbitrage is feasible. Example:

EXAMPLE

Assume the following information:

- You have £800 000 to invest.
- The current spot rate of the dollar is £0.625.
- The 90-day forward rate of the pound is £0.625.
- The 90-day interest rate in the United Kingdom is 2%.
- The 90-day interest rate in the United States is 4%.

These interest rates are rather high and are for illustrative purposes only. 90 days is 90/360 or ¼ of a year so the annual rate in the US is approximately 16% (i.e. 4 × 4%)!

Based on this information, you should proceed as follows:

1. On day 1, convert the £800 000 to $1 280 000 and deposit the $1 280 000 in a US bank.
2. On day 1, sell $1 331 200 at the 90-days forward rate at the contract rate of £0.625 to the dollar. Note that by the time the deposit matures, you will have $1 280 000 × 1.04 = $1 331 200 (including interest).
3. In 90 days when the deposit matures, you can fulfil your forward contract obligation by converting your $1 331 200 into £832 000 (based on the forward contract rate of £0.625 per dollar).

This reflects a 4% return over the three-month period (calculated as (£832 000 − £800 000) / £800 000 = 4%), which is twice as much as the 2% return on the UK investment over a similar period. The return on the foreign deposit is known on day 1, since you know when you make the deposit exactly how many dollars you will get back from your 90-day investment.

This strategy would not be advantageous if it earned 2% per annum or less, since you could earn 2%on a domestic deposit for the same level of (no) risk.

Interest Rate Parity (IRP)

There is an equilibrium state referred to as **interest rate parity** (IRP) once market forces cause interest rates and exchange rates to adjust such that covered interest arbitrage is no longer feasible. In equilibrium, the forward rate differs from the spot rate by a sufficient amount to offset the interest rate differential between two currencies.

Purchasing Power Parity (PPP)

When the inflation rate in country A (for example) rises, the demand for its currency declines as its exports decline (due to its higher prices). In addition, consumers and firms in country A tend to increase their importing (increasing the supply of home currency for foreign currency). The reduced demand for country A's currency and the increased supply of country A's home currency for foreign currency both place downward pressure on A's currency. **Purchasing Power Parity** bases its predictions of exchange rate movement on changing patterns of trade due to different inflation rates between countries. *Absolute PPP* states that without international trade barriers and transport costs, consumers shift their demand to wherever prices are lower, so prices of the same basket of products in two different countries should be equal when measured in a common currency. *Relative PPP* accounts for the possibility of market imperfections such as transportation costs, tariffs, and quotas. However, the rate of change in the prices of the baskets should be similar when measured in a common currency.